

Guidance for CoBit bitstream diffusion

What we ran, what we found, and what changed

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Summary in five lines.

1. Under deterministic sampling, all three guidance methods help: CFG +0.081, AutoGuidance +0.066, Self-Guidance +0.048.
 2. Self-Guidance is *not* a numerical-solver artefact — a correctly implemented second-order solver (Heun) gives nothing at matched cost.
 3. Switching on stochastic churn — a sampler flag costing *no extra forward passes* — raises the plain baseline by +0.120, more than any guidance method delivers.
 4. Under churn, CFG survives but must be re-tuned ($w=12 \rightarrow 2$); AutoGuidance reverses; Self-Guidance reverses at every scale tested.
 5. Guidance and majority voting compose *additively*, but at equal forward passes extra samples remain the better buy.
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1 Setup

CoBit is a continuous diffusion language model over *bitstreams*: it denoises a real-valued vector whose coordinates are bit probabilities, and decodes to tokens at the end. We evaluate on GSM8K (grade-school maths word problems), zero-shot from a TinyGSM-trained checkpoint. The model writes a short Python program; we execute it in a sandbox and compare the returned number to the gold answer. The metric throughout is **exact match**.

Three guidance mechanisms are compared. Writing D for the model’s predicted clean bit-probabilities:

$$\underbrace{D_u + w(D_c - D_u)}_{\text{CFG}} \quad \underbrace{D_{\text{bad}} + w(D_{\text{good}} - D_{\text{bad}})}_{\text{AutoGuidance}} \quad \underbrace{D + w(D_t - D_{\text{prev}}) \cdot \frac{\delta}{\Delta \log \sigma}}_{\text{Self-Guidance}}$$

CFG contrasts a conditional against an unconditional (null-prompt) branch; AutoGuidance contrasts the final checkpoint against a deliberately weaker earlier one; Self-Guidance contrasts predictions at adjacent noise levels. CFG and AutoGuidance each need *two* network evaluations per step; Self-Guidance reuses a cached prediction and is free.

Statistics. Screening runs use a fixed 250-problem prefix and one seed. Every number quoted below is **confirmation grade**: all 1319 GSM8K test problems, three seeds, with per-problem paired bootstrap intervals (20,000 resamples). Pairing removes problem-difficulty variance, which is what makes 2-point effects resolvable at this sample size.

The parameter γ controls EDM-style *stochastic churn*: at each step the state is partially re-noised before being denoised again, with $s_{\text{churn}} = \gamma(\text{NFE} - 1)$, capped at $\sqrt{2} - 1$. Crucially this adds *no* network evaluations — measured at 256 NFE, ~ 94 s and 2.31 GB for every γ we tested.

2 What we ran

Stage	Experiment	Cells	Question
<i>Guidance at $\gamma = 0$</i>	cfg_coarse, cfg_high	18	CFG scale, 0.25–30
	ag, ag_high, ag_ema	50	AutoGuidance badness $\times$ scale
	sg, sg_delta, sg_mf	52	Self-Guidance variant, δ , matched filter
	cfg/ag/sg_confirm	36	full-size confirmation, 3 seeds
	factorial_confirm	36	all 12 method combinations
	nfe	28	quality vs compute, 8–512 steps
	null_ablation	3	sensitivity to the CFG null
<i>Controls</i>	solver_control	15	is Self-Guidance just 2nd-order integration?
	compute_control	4	disjoint seeds for majority voting
	repro	4	did the evaluation change move any number?
<i>Stochasticity</i>	stoch_screen	20	γ screen, 4 methods
	stoch_confirm	36	full-size confirmation
	churn_anatomy	21	re-tune CFG at $\gamma=0.3$; can SG be rescued?
total		329	

Two analyses cost *no* GPU time because the per-problem executed answers were already recorded: the compute-matched majority-vote controls (§5) and the guidance $\times$ voting composition test (§6).

3 Guidance under deterministic sampling

Each method at its own tuned operating point, against a single baseline sample.

Configuration	Exact match	Δ	95% CI	NFE
CFG $w=12$	0.2196	+0.0811	[+0.0675, +0.0953]	1024
AutoGuidance $w=15$	0.2047	+0.0662	[+0.0533, +0.0794]	1024
SG-prev $w=2$	0.1868	+0.0483	[+0.0392, +0.0576]	512
SG-exact $w=1$	0.1233	−0.0152	[−0.0220, −0.0083]	1024
baseline	0.1385	—	—	512

Two details matter more than the ranking. **SG-prev runs at the baseline’s cost** — it is the only free method. And **SG-exact, which costs twice as much, is worse than doing nothing**: the cheap approximation is not approximating the expensive one.

Combinations destroy the model. The full $2 \times 2 \times 3$ factorial (Fig. 1) shows every combination is worse than its best component, and CFG+AutoGuidance collapses to **0.0005** — essentially no correct answers. The interaction is -0.166 . Diagnostically the guidance-to-score ratio rises from 0.20 (CFG alone) to 0.86 (CFG+AG) to 11.6 (three-way): stacking multiplies the amplification and drives every bit to a hard 0 or 1.

4 Is Self-Guidance really guidance?

SG-prev’s direction is a difference between consecutive predictions, divided by the log- σ spacing — an amplified finite-difference derivative. That is structurally what a second-order ODE

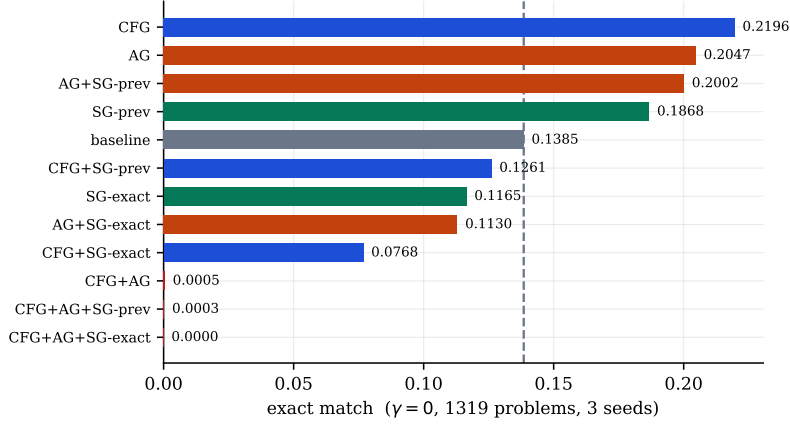


Figure 1: All twelve factorial conditions, $\gamma = 0$. Red bars are the collapsed combinations. Dashed line is the unguided baseline.

solver computes, from the same two quantities, so “guidance” and “better numerical integration” predicted the same $+0.048$.

We implemented Heun (a genuine second-order solver) and compared at *measured* forward passes. This mattered: the recorded NFE field counted only guidance branches and was blind to solver order, under-reporting Heun by $2\times$. Instrumenting `model.forward` gave DDIM exactly 1.00 evaluations/step and Heun $2N-1$.

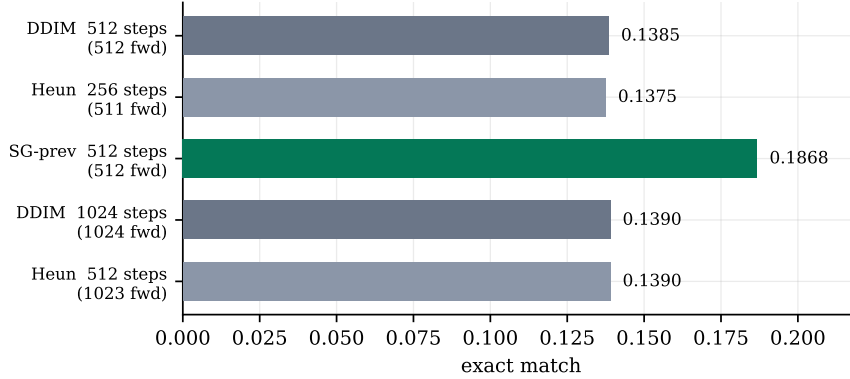


Figure 2: Solver control at matched forward passes. Heun matches DDIM at both budgets; SG-prev does not.

SG-prev beats Heun by $+0.0493$ $[+0.0402, +0.0586]$ at identical cost, while Heun differs from DDIM by -0.0010 $[-0.0030, +0.0010]$ (not significant). Doubling DDIM’s steps also does nothing (0.1390 vs 0.1385). **The solver hypothesis is rejected.** What SG-prev encodes is still unknown.

5 Stochastic churn changes everything

Every result above was measured with churn disabled. It should not have been held fixed.

The unguided baseline gains $+0.1200$ $[+0.1056, +0.1350]$ going from $\gamma=0$ to $\gamma=0.3$ — larger than any guidance effect in this study, at no additional cost. Measured against the *same- γ* baseline, the guidance methods behave very differently:

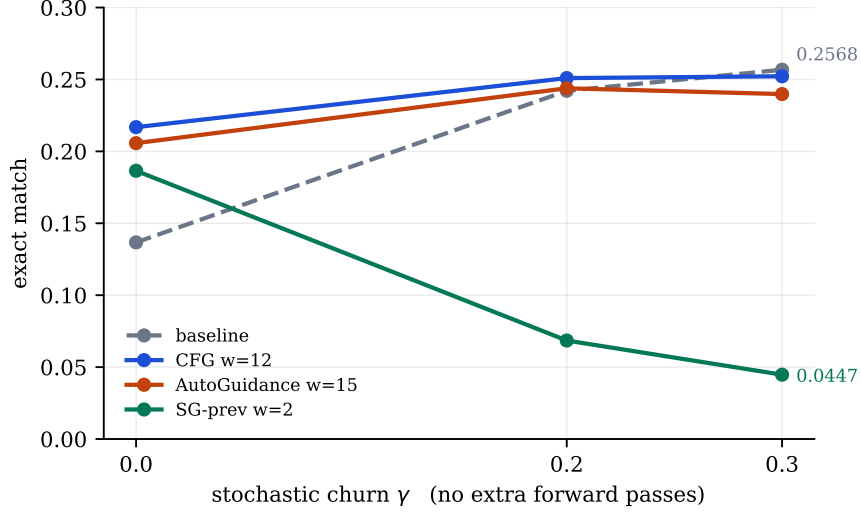


Figure 3: Exact match against churn γ . The unguided baseline (dashed) climbs past every guided line. SG-prev inverts completely. 1319 problems, 3 seeds.

vs. same- γ baseline	$\gamma = 0$	$\gamma = 0.2$	$\gamma = 0.3$
CFG $w=12$	+0.0801	+0.0088 n.s.	-0.0045 n.s.
AutoGuidance $w=15$	+0.0690	+0.0018 n.s.	-0.0169
SG-prev $w=2$	+0.0498	-0.1736	-0.2120

But CFG was mis-tuned, not superseded. $w=12$ was chosen at $\gamma=0$. Sweeping the scale at $\gamma=0.3$ separates “guidance stops working” from “guidance is tuned for the wrong regime”. It is the second: $w=2$ gives 0.2757, +0.0190 [+0.0068, +0.0316] over the plain stochastic baseline.

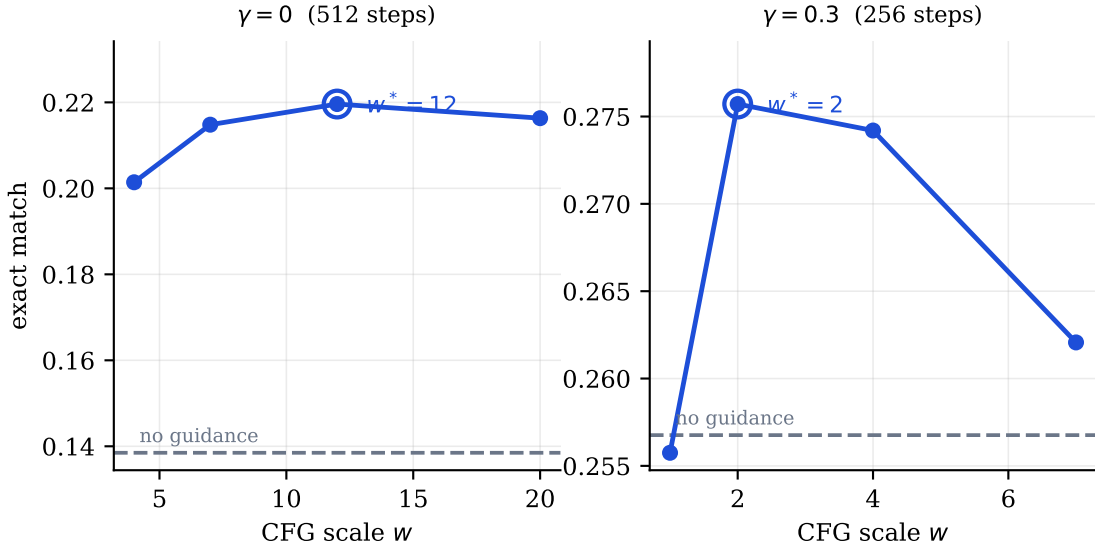


Figure 4: The optimal CFG scale moves from $w^* \approx 12$ to $w^* \approx 2$ when churn is switched on, and the gain shrinks from +0.080 to +0.019. **A guidance scale quoted without its sampler regime is under-specified.**

Self-Guidance cannot be rescued. If the direction were right and only the magnitude wrong, shrinking the scale should restore it. It does not: at $\gamma=0.3$, $w = 0.06/0.125/0.25$ give -0.0131/-

0.0286/−0.0551 — still harmful $33\times$ below the $\gamma=0$ optimum, and harm *linear in w* , the signature of a systematically wrong direction rather than amplified noise. Two mechanisms are already excluded by the traces: the spacing normaliser is unchanged across γ , and $\log \sigma$ remains monotone. Working hypothesis (untested): churn re-noises the state each step, so the cached prediction comes from a strictly noisier point and the difference acquires a consistent backward component.

6 Compute: is guidance worth its extra evaluations?

CFG and AutoGuidance double the network evaluations per sample. The honest comparison spends those same evaluations on *more samples* and combines them by majority vote — deployable, unlike the pass@ k oracle bound.

Arm ($\gamma = 0.3$)	NFE	Samples	Exact match
baseline, single sample	256	1	0.2568
baseline maj@2	512	2	0.2881
baseline maj@3	768	3	0.3146
CFG $w=2$, single	512	1	0.2757
CFG $w=2$ maj@2	1024	2	0.3060

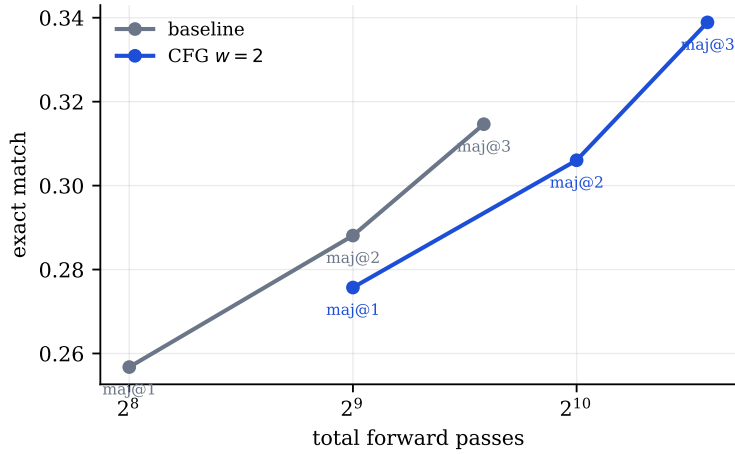


Figure 5: Guidance and voting compose additively: voting adds +0.0313 to the baseline and +0.0303 to CFG — statistically the same increment, so the two mechanisms are independent.

The verdict depends on what is held fixed, and these must not be collapsed into one “compute” number:

Held fixed	Comparison	Δ	Verdict
number of samples (latency-bound)	CFG maj@2 – baseline maj@2	+0.0179	guidance wins
	[+0.0023, +0.0339]		
forward passes	CFG maj@2 (1024) – baseline maj@3 (768)	−0.0086	inconclusive*
	[−0.0258, +0.0088]		

*and the baseline used 25 % *less* compute to achieve it.

For a latency-bound single-stream deployment, guidance is worth it. For a throughput-bound one, extra samples are the better buy. Fig. 6 shows the same story at $\gamma = 0$ across the NFE range.

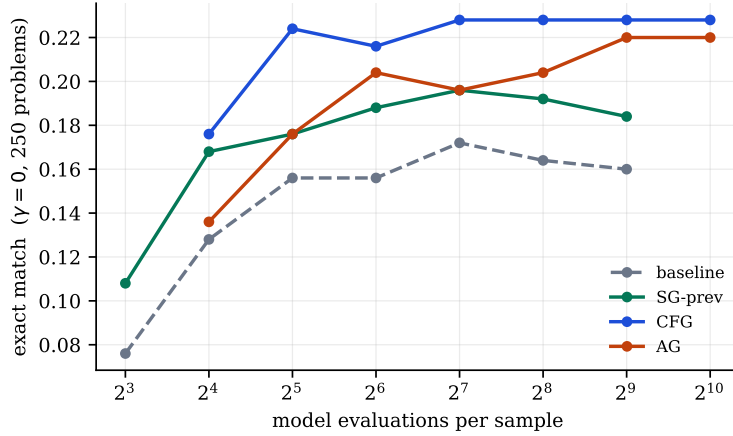


Figure 6: Quality against model evaluations at $\gamma = 0$ (250-problem screen). The baseline plateaus near 0.17; guidance buys a higher plateau, not a faster one.

7 What is established, and what is not

Claim	$\gamma = 0$	$\gamma = 0.3$	Status
CFG helps	+0.0811	+0.0190 ($w=2$)	survives, re-tuned
AutoGuidance helps	+0.0690	-0.0169	reverses
SG-prev helps	+0.0498	-0.2120	reverses at every scale
SG-prev is a solver effect	—	—	rejected
Guidance beats equal-compute voting	+0.0325	-0.0124	only at $\gamma = 0$
Combining methods fails	0.0005	untested	unchanged

Best known configuration: unguided baseline at $\gamma=0.3$ with two samples and a majority vote — 0.2881 at 512 NFE, no guidance at all. Best single-stream option: CFG $w=2$ at $\gamma=0.3$ (0.2757). For comparison, what the first version of this study recommended — CFG $w=12$ at $\gamma=0$ — scores 0.2168.

Not established. (i) *Why churn works.* The single largest effect we found is a free sampler flag and its mechanism is unexplained. (ii) *Why SG-prev’s direction is wrong under churn;* the re-noising hypothesis is untested. (iii) *Whether the factorial collapse persists under churn* — it was measured only at $\gamma = 0$. (iv) *Anything beyond this task:* one model, one dataset, one sampler family; AutoGuidance and Self-Guidance are implemented for the DDIM sampler only. (v) Effects below ~ 2 points are not resolvable with 1319 binary outcomes.

Methodological checks. Re-running four headline cells under the current code reproduced the originals *bit-for-bit per problem*, so an intervening change to the evaluation loop moved nothing. Separately, the 250-problem screening set is a prefix of the 1319-problem confirmation set, so 19 % of the confirmation data helped choose what was confirmed on it; re-estimating on the disjoint 1069 problems shifts every effect by ≤ 0.002 .

8 Next

In priority order: (1) explain the churn effect — trajectory-level analysis of when it acts and whether it improves individual trajectories or only diversifies them; (2) test the re-noising hypothesis for Self-Guidance with a matched counterfactual; (3) measure guidance geometry (direction vs magnitude) under churn; (4) re-test the factorial collapse at $\gamma > 0$. Items (1) and (3) need trajectory instrumentation that does not yet exist — intermediate answer decoding and pairwise guidance cosines — which is code, not GPU time.

Budget: **107 of** ~ 1000 **A100-hours** spent. Nothing is currently queued.

329 recorded runs on CSD3 Ampere, branch `tasks/fkc-temperature`. Every figure and number in this document regenerates from the per-problem outcome vectors in `runs/guidance/` via `docs/report/make_figures.py`; nothing is hand-written. Each run records commit, config, checkpoint, sampler, schedule, NFE, guidance scales, seed, GPU, SLURM job id, wall-clock and peak memory.